Corporate Finance

Lecture 7: Capital Structure in a Perfect Market (I)

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Plan

- Recap from lecture 6
- Ask whether financing a project with debt or with equity changes what it is worth
- Follow one entrepreneur who finances the same project two ways, and see why debt makes the equity riskier
- Meet Modigliani and Miller's first proposition, and the arbitrage behind it
- Read a market value balance sheet and follow a leveraged recapitalization

Mostly based on the BDM textbook Chap. 14.1-14.2.

Recap lecture 6

Recap: the investment decision

A project is worth doing when its **expected return** is at least its **cost of capital**: the return investors could earn elsewhere at the same risk.

- The money for a project comes from **equity** (selling a share of the firm), from **debt** (borrowing), or from a **mix** of the two.
- Each source has its own cost, because each investor bears a different risk. Lecture 6 was about measuring these costs:
  - the cost of equity r_E, with the CAPM;
  - the cost of debt r_D: the yield minus the expected loss from default, or the CAPM with a debt beta;
  - their combination for the whole firm or a project: r_U and the WACC.

Recap: the cost of equity

The CAPM: what shareholders require for the market risk they bear (Chap. 12.1).

$$r_E = r_f + \beta_E \left(E[R_M] - r_f \right)$$

- **Each component**, and where it comes from (Chap. 12.2-12.3):
  - **The market portfolio** R_M: all stocks weighted by market value; in practice an index.
  - **The risk-free rate** r_f: long government bond yields.
  - **The market risk premium** $E[R_M] - r_f$: about 3-5%, from past excess returns or from dividend yield plus growth.
  - **The beta** β_E: the slope of the stock's excess returns against the market's, from a few years of data.

Recap: the cost of debt

The yield to maturity is a **promised** return. When the borrower may default, the **expected** return is lower (Chap. 12.4):

$$r_D = \text{YTM} - p \times L$$

- p is the probability of default and L the loss rate when it happens. For a safe borrower p is close to zero and r_D is the yield.
- **Second route: a debt beta.** Use the CAPM, $r_D = r_f + \beta_D \times$ premium, with average betas by rating: near zero for investment grade, up to 0.3 for the riskiest bonds.

Recap: a project's cost of capital

A project has no beta of its own: borrow it from a listed **comparable** in the same business (Chap. 12.5).

- **Comparable without debt**: its equity beta is the project beta. Done.
- **Comparable with debt**: its shares are riskier than its assets, so first **unlever**. The assets belong to shareholders and lenders together, so their cost of capital is the weighted average of the two:

$$r_U = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D \qquad \beta_U = \frac{E}{E+D}\, \beta_E + \frac{D}{E+D}\, \beta_D$$

- Cash counts as **negative debt**: use net debt = debt - cash for D.

Recap: the WACC

When a firm borrows, interest is tax deductible, so the effective cost of debt is $r_D(1-\tau_C)$ and the **weighted average cost of capital** is (Chap. 12.6)

$$r_{WACC} = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D\,(1-\tau_C)$$

- τ_C is the corporate tax rate. Without the tax term the formula is the unlevered cost of capital r_U of the previous slide, also called the **pre-tax WACC**.

Beginning of lecture 7 content

Lecture 6 measured the cost of equity and the cost of debt, and debt looked cheaper. Today we ask whether borrowing more therefore makes a firm worth more.

What CFOs think matters most

- In a survey of CFOs (Servaes and Tufano, 2006), **capital structure** ranks first among the finance functions that add value to the firm, ahead of investment decisions and risk management.
- **Capital structure** is the mix of debt and equity a firm uses to finance its assets.

Modigliani and Miller

- Franco Modigliani (1918-2003), Nobel prize 1985, and Merton H. Miller (1923-2000), Nobel prize 1990.
- "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review*, 1958.
- Their two propositions tell us what leverage does to the value of the firm (today) and to its cost of capital (Lecture 8).

What the MM theorems give us

- The theorems of Modigliani and Miller (MM) identify the effect of leverage on
  - the value of the firm (today, Proposition I),
  - the cost of equity (Lecture 8, Proposition II),
  - the cost of capital (Lecture 8).
- We will use MM to
  - optimize the capital structure,
  - take investment decisions: which discount rate to use for a project,
  - understand how financing decisions can change the value of the firm.

Outline

A. Financing a project with equity (Chap. 14.1)

B. Adding debt: levered equity and a common fallacy (Chap. 14.1)

C. Perfect markets and Modigliani-Miller Proposition I (Chap. 14.2)

D. Homemade leverage and arbitrage (Chap. 14.2)

E. The market value balance sheet (Chap. 14.2)

Introduction

- Should you finance new projects with debt or equity?
- What is the optimal capital structure for a company?
- We will begin by assuming that there are no taxes and no transaction costs (and that markets are perfect).

The Modigliani-Miller conclusion

- Modigliani and Miller asked a simple question: in a **perfect capital market** (no taxes, no transaction costs, fair prices), does the way a firm finances itself change what it is worth?

MM Proposition I In a perfect capital market, the total value of a firm's securities is equal to the market value of the total cash flows generated by its assets, and is not affected by its choice of capital structure.

- **Why:** the firm's assets produce the same cash flows however they are financed. Debt and equity only decide **who receives** those cash flows: lenders, shareholders, or both. The total to share stays the same.
- **How we will see it:** we follow one project, first financed with equity only (section A), then partly with debt (section B), and compare its value in the two cases. Section C states the result in general.

No taxes: what it does to the WACC

- MM I assumes **no taxes**. The WACC has a tax term on the debt because interest is tax deductible:

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_C)$$

- With a corporate tax rate $\tau_C = 0$, the factor $(1 - \tau_C)$ equals 1 and disappears:

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

A. Financing a project with equity only

- An entrepreneur (call the firm Tour moche SA) has a project: invest **800** today; next year it pays **1400** if the economy is strong or **900** if it is weak. Both outcomes are equally likely.

Year 0: investment	Year 1: strong (prob. 0.5)	Year 1: weak (prob. 0.5)
-800	1400	900

- We are **given the cost of equity: 15%**, the 5% risk-free rate plus a 10% risk premium (in practice it comes from the CAPM, Lecture 6).
- The key question: **how much is the firm worth?**
- The price of the firm is the Present Value of its activities: we discount the future cash flows using the cost of capital.

The cost of capital with equity only

- The cost of capital in a project financed only with equity, is just the cost of equity

$$\text{cost of capital} = \text{cost of equity} = 15\%$$

Step 1: what is the firm worth? (its fair price)

- The value of a firm is equal to the value of its activites: Tour moche S.A. has only this project, so its value equals the value of the project.
- Expected cash flow next year: $0.5 \times 1400 + 0.5 \times 900 = 1150$.
- Discounted at 15% cost of capital:

$$\text{PV} = \frac{1150}{1.15} = 1000$$

- So the **value of the firm** is 1000.

Unlevered firm	Year 0	Strong	Weak
Debt	0	0	0
Levered equity	1000	1400	900
Firm	**1000**	**1400**	**900**

what is the firm worth? (its fair price)

- The entrepreneur can sell the firm at 1000, pay for the 800 investment and keep **keep 200**

$$\text{NPV} = -800 + 1000 = 200$$

- Equity in a firm with no debt is called **unlevered equity**.
- As we computed the price using the 15% discount rate (the cost of capital), by construction the returns of the shareholders that buy the firm will be 15%.

B. The same project, financed another way

- Same project.
- Now the entrepreneur **borrows 500** and sells equity for the rest.
- She can borrow at the risk-free rate 5%: even in a weak economy the project pays 900, enough to repay. The loan is **risk free**.
- Next year the firm will owe the lenders $500 \times 1.05 = 525$, whatever the economy does:

Levered firm	Year 0	Strong	Weak
Debt	500	**525**	**525**
Levered equity	$E = ?$	?	?
Firm	?	?	?

- *What is left for the shareholders?*

Shareholders part

- Lenders must be paid **before** shareholders receive anything. Shareholders get what remains:
  - strong economy: $1400 - 525 = 875$
  - weak economy: $900 - 525 = 375$
- Equity in a firm that also has debt is called **levered equity**.

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	$E = ?$	**875**	**375**
Firm	?	?	?

Firm value

- Put the lenders and the shareholders together:
  - strong economy: $525 + 875 = 1400$
  - weak economy: $525 + 375 = 900$
- Together they receive **exactly the project's cash flows**:

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	$E = ?$	875	375
Firm	?	**1400**	**900**

- These are the same values tan for the unlevered firm!!!!! Nothing is created and nothing is lost.
- Now, let's evaluate the value of the firm and the value of equity...

The value of the firm and the Law of One Price

- We just showed that the cash flows of the firm are the sam under both scenarios.
- **The Law of One Price**: two investments that pay the same cash flows in every state of the economy must sell for the same price.
- As a consequence, the value of the firm will be the same: 1000.

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	??	875	375
Firm	**1000**	1400	900

The value of equity

- We have the value of the firm and the debt, we can derive the equity with the identity:

$$D + E = 1000 \qquad \Rightarrow \qquad E = 1000 - 500 = 500$$

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	**500**	875	375
Firm	**1000**	1400	900

Is the entrepreneur worse off?

	Equity only	Debt and equity
Sold to shareholders	1000	500
Borrowed from lenders	0	500
Total raised	**1000**	**1000**
Invested	800	800
Kept (the NPV)	**200**	**200**

- She raises the same 1000 and keeps the same 200.

The entrepreneur is **indifferent** between the two ways of financing the project.

The common fallacy: a tempting shortcut

Before Modigliani and Miller, many people reasoned like this:

1. **What do shareholders expect next year?** In the strong state they get 875, in the weak one 375. On average:

$$0.5 \times 875 + 0.5 \times 375 = 625$$

2. **What is that worth today?** Discount it at 15%, the rate we used for the firm without debt:

$$625/1.15 = 543$$

3. **Conclusion:** the equity is worth 543, more than the 500 we found with the Law of One Price. So borrowing would create value out of nothing.

This contradicts the Law of One Price

The common fallacy: a tempting shortcut

- The error consist in using the same dicount rate of 15% for the equity in the levered case.
- In the levered case, shareholders pay **500** today, so they actual returns are:
  - strong economy: they receive 875, a return of $875/500 - 1 = +75\%$
  - weak economy: they receive 375, a return of $375/500 - 1 = -25\%$
- Expected return: $0.5 \times 75\% + 0.5 \times (-25\%) = 25\%$.
- Compare with the equity of the firm without debt (section A):

Equity	Pays today	Strong	Weak	Expected return of shareholders
Without debt	1000	$+40\%$	-10%	15%
With debt of	500	$+75\%$	-25%	**25%**

Why 25%?

- The economy is the same in both cases. What changed is how much the shareholders' return **moves** with it:
  - without debt, the return goes from -10% to $+40\%$: a swing of **50 points**;
  - with debt, from -25% to $+75\%$: a swing of **100 points**.
- Levered equity is **riskier**, so investors require a higher expected return. 15% was the price of the first risk; 25% is the price of the second.
- **The shortcut's mistake was the discount rate**: 15% is the rate for unlevered risk. Discount at the right rate and the puzzle disappears:

$$625/1.25 = 500$$

Cheap debt, dearer equity

- Looked at on its own, debt is cheap: 5% against 25% for the levered equity.
- But the firm uses both. Its **average cost of capital**, weighting each source by its value, is

$$\frac{500}{1000} \times 5\% + \frac{500}{1000} \times 25\% = 15\%$$

- The same 15% as with equity alone.

Exercise: borrow only 200

- Same project, same entrepreneur. This time she borrows only **200** at 5% and sells equity for the rest.
- According to Modigliani and Miller, what is the equity worth today?
- What do the shareholders receive next year, in a strong and in a weak economy?
- What return do they earn in each case, and what is their expected return?
- What is the firm's average cost of capital?

Exercise: borrow only 200 (con't)

- **Step 1, the equity value.** Debt plus equity is still worth 1000, so $E = 1000 - 200 = 800$.
- **Step 2, the cash flows.** The firm owes $200 \times 1.05 = 210$. Shareholders get $1400 - 210 = 1190$ or $900 - 210 = 690$.
- **Step 3, the returns.** On 800 paid today: $1190/800 - 1 = +48.75\%$ or $690/800 - 1 = -13.75\%$. Expected: $0.5 \times 48.75\% + 0.5 \times (-13.75\%) = 17.5\%$.
- **Step 4, the average cost of capital.** $0.8 \times 17.5\% + 0.2 \times 5\% = 15\%$. Again.
- Less debt than before, so the equity is less risky than with 500 of debt (17.5% against 25%), but riskier than with none (15%).

What section B showed

1. Lenders are paid first; shareholders get what remains.
2. Debt and equity together receive **the project's cash flows**.
3. By the Law of One Price, debt and equity together are worth **what the project is worth**: 1000, whatever the mix. The entrepreneur is indifferent.
4. Shareholders of a firm with debt earn a **higher expected return**, because they bear **more risk**.
5. The **average cost of capital** does not move: cheap debt is offset by dearer equity.

Next: Modigliani and Miller show this is a general result, not a feature of our example.

C. What is a perfect capital market?

- Our example made hidden assumptions. Modigliani and Miller (1958) spell them out; together they define a **perfect capital market**:

  1. **Investors and firms trade the same securities at fair prices**, equal to the present value of their cash flows.
  2. **No taxes, transaction costs or issuance costs.** (Nothing leaked out when she sold the equity or took the loan.)
  3. **No bankruptcy costs.** Default itself destroys no cash flow; lenders simply receive what is left.
  4. **All investors can borrow and lend without restrictions at the risk-free rate.** (Our entrepreneur borrowed at the same 5% as anyone else.)
  5. **Financing does not change the project's cash flows**, and does not reveal new information about them.

- Remove any one of them and the conclusion can change. That is the subject of the next lectures.

MM Proposition I

MM Proposition I. In a perfect capital market, the total value of a firm's securities is equal to the market value of the total cash flows generated by its assets, and is not affected by its choice of capital structure.

- In plain words: **the pie is the same size however you slice it.** The project's cash flows are the pie; debt and equity are the slices.
- In our example the pie was worth 1000 with no debt, with 200 of debt, and with 500 of debt. Only the slices changed.

Why MM I holds: two ways to see it

- **The Law of One Price.** Without taxes or transaction costs, everything the firm's assets produce is paid to its security holders, and nothing else is. Same total cash flows, so same total value, whatever the mix of securities.
- Miller's version: the pizza delivery man asks Yogi Berra whether he wants his pizza cut into four or eight slices. Yogi: "Cut it in eight pieces, I'm feeling hungry tonight." Cutting the pizza differently does not make more pizza.

MM and the real world

- Students often ask why MM matters if real markets are not perfect.
- Galileo showed that **without friction**, all bodies fall at the same speed. Drop a feather and a stone and it does not hold. The reason is not that Galileo was wrong, but that **air friction** slows some objects more than others.
- MM is the same kind of result. In practice capital structure **does** affect value. MM tells us that any such effect must come from a **friction** in the capital market: taxes, bankruptcy costs, information problems.

D. Homemade leverage

- Suppose investors would prefer a different capital structure from the one the firm has chosen.
- The homemade leverage is the mechanism in which the investor combines the company shares with other assets to reach the leverage level it wants.
- Condition: investors can borrow at the same rate as the firm.

Increase the leverage yourself

- Let's assume that the investor wants more leverage: he takes a loan of 500 and buys the shares of the unlevered Tour Moche S.A.

Portfolio	Year 0	Strong	Weak
Buy unlevered equity	-1000	$+1400$	$+900$
Borrow 500 (margin loan)	$+500$	-525	-525
Total	-500	$+875$	$+375$

- The portfolio pays **exactly what levered equity pays**, for a cost of 500. By the Law of One Price, levered equity must also be worth 500.

Undo the firm's leverage

- Now the firm has borrowed 500, but the investor would prefer unlevered equity. He buys **both** the debt and the levered equity.
- That way the debt he owns neutralizes the one of the firm.

Portfolio	Year 0: cost	Strong	Weak
Debt	500	525	525
Levered equity	500	875	375
Total	1000	1400	900

- Same cash flows as unlevered equity, for 1000.

Either way, the investor picks the leverage he wants. The firm's choice changes nothing for him, so he will not pay more for one structure than for the other.

Proof of MM I with homemade leverage and the law of one price

- We already showed that the firm has the same value independently of the capital structure.
- Now we will reinforce that conclusion: given the homemade leverage and the law of one price, the can't have different firm value.
- Take two firms with **the same assets**, paying 1400 or 900 next year:
  - **U** has no debt. Its shares are worth V_U.
  - **L** has borrowed 500 at 5%. Its debt is worth $D = 500$ and its shares E_L. The firm is worth $V_L = D + E_L$.
- **Claim (MM I):** $V_U = V_L$. The two firms must sell for the same price.
- **How we prove it.** Suppose the prices were **different**. Using the two homemade-leverage tricks, an investor could pocket

Proof of MM I (con't): U is too expensive

- Suppose **U sells for 1050** while L's debt and shares sell for $500 + 500 = 1000$. An investor who owns U's shares **undoes** L's leverage (slide 40): he sells U and buys all of L's debt and all of L's shares.

Trade	Year 0	Strong	Weak
Sell U's shares	$+1050$	-1400	-900
Buy L's debt	-500	$+525$	$+525$
Buy L's shares	-500	$+875$	$+375$
Total	$+50$	0	0

- Next year nothing changes for him: he gave up 1400 or 900 and receives exactly 1400 or 900. Today he earned **50 today**.
- Every investor sells U and buys L. U's price falls, L's rises, until $V_U = D + E_L$.

Proof of MM I (con't 2): U is too cheap

- Now suppose **U sells for 990** while L's shares sell for 510, so $V_L = 500 + 510 = 1010$. An investor who owns L's shares **makes his own leverage** (slide 39): he sells L's shares, borrows 500 himself and buys U.

Trade	Year 0	Strong	Weak
Sell L's shares	$+510$	-875	-375
Borrow 500 at 5%	$+500$	-525	-525
Buy U's shares	-990	$+1400$	$+900$
Total	$+20$	0	0

- Same cash flows next year, **20** in his pocket today. Everyone sells L and buys U until $V_U = D + E_L$ again.

Both directions lead to the same place: by the **law of one price**, $V_U = V_L = D + E_L$. Firm value does not depend on the debt. This is MM I.

E. The market value balance sheet

According to MM1, the **market value** of the firm's assets is equal to the **market value** of the securities issued by the firm.

$$\text{Equity} = \text{Assets} - \text{Debt and other liabilities.}$$

Assets	Liabilities
Collection of Assets and Investments Undertaken by the Firm:	Collection of Securities Issued by the Firm:
Tangible Assets	Debt
Cash	Short-term debt
Plant, property, and equipment	Long-term debt
Inventory and other working capital (and so on)	Convertible debt
Intangible Assets	Equity
Intellectual property	Common stock
Reputation	Preferred stock
Human capital (and so on)	Warrants (options)
Total Market Value of Firm Assets	**Total Market Value of Firm Securities**

What to remember

- **MM Proposition I.** In a perfect capital market the total value of a firm's securities equals the value of its assets, whatever the capital structure.
- **Perfect market** means: fair prices; no taxes, transaction or issuance costs; no bankruptcy costs; everyone borrows and lends at the risk-free rate; financing changes neither the investment cash flows nor the information about them.
- **Leverage makes equity riskier**, greater leverage increases the risk for shareholders, as the debt has payment priority. They will demand grater compensation which raises the cost of equity.
- **Homemade leverage**: investors can borrow or lend to get the leverage they want, so the firm's choice gives them nothing extra.

Quiz

- Why is the levered equity of Tour moche SA worth 500 and not 543?
- Name the five conditions of a perfect capital market.
- The entrepreneur borrows 750 instead of 500. What is the value of the equity, what are its cash flows, and what is its expected return?

Quiz: answers

- **500, not 543.** The 543 discounts the expected levered cash flow of 625 at the unlevered rate of 15%. Levered equity is riskier and requires 25%: $625/1.25 = 500$. The 500 also follows from the Law of One Price: $1000 - 500$.
- **Perfect market.** Fair prices for all securities; no taxes, transaction or issuance costs; no bankruptcy costs; unrestricted borrowing and lending at the risk-free rate; financing does not change the investment cash flows nor reveal information about them.
- **Borrow 750.** Equity is worth $1000 - 750 = 250$. The firm owes $750 \times 1.05 = 787.5$, so equity receives $1400 - 787.5 = 612.5$ or $900 - 787.5 = 112.5$: returns $+145\%$ or -55%, expected **45%**. Risk premium 40%, four times the unlevered premium, for four times the sensitivity.